COMPUTER ORGANIZATION and ARCHITECTURE

Unit 1-A

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Outline

Digital Computers

- Introduction, Block diagram of Digital Computer
- Definition of Computer Organization, Computer Design and Computer Architecture

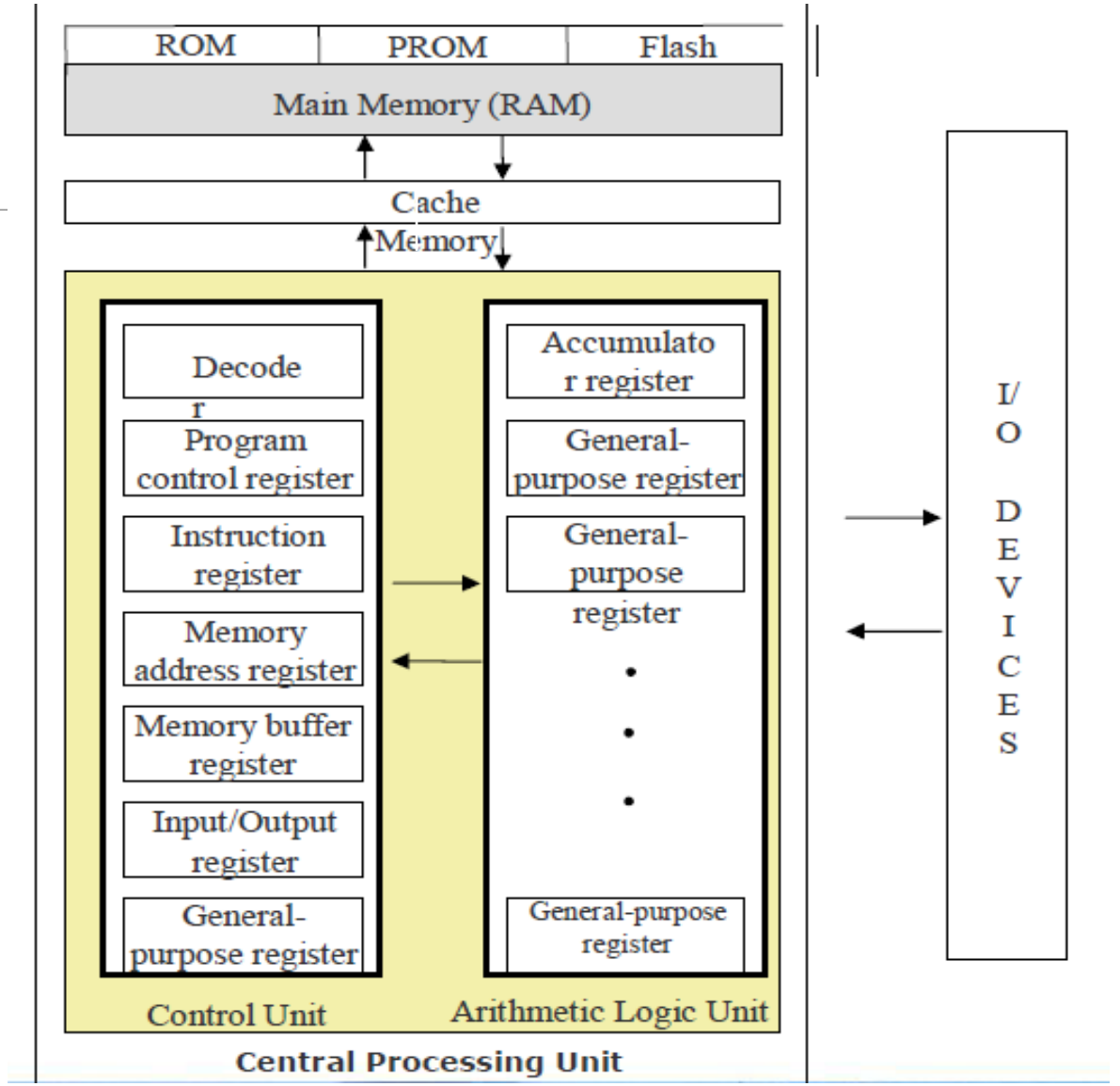
Register Transfer Language and Micro operations

- Register Transfer language, Register Transfer, Bus and memory transfers
- Arithmetic Micro operations, logic micro operations, shift micro operations
- Arithmetic logic shift unit

Basic Computer Organization and Design

- Instruction codes, Computer Registers Computer instructions
- Timing and Control, Instruction cycle, Memory Reference Instructions
- Input - Output and Interrupt

Components of CPU



CPU – Central processing Unit

The core of a computer system

Performs all major calculations and comparisons

Activates and controls the operations of other units of a computer system

Three basic components are

- Control Unit (CU)
- Arithmetic Logic Unit (ALU)
- Registers

Control Unit

- One of the two basic components of CPU
- Acts as the central nervous system of a computer system
- Selects and interprets program instructions, and coordinates execution
- Has some special purpose registers and a decoder to perform these activities

Arithmetic logic unit

- One of the two basic components of CPU.
- Actual execution of instructions takes place in ALU
- Has some special purpose registers
- Has necessary circuitry to carry out all the arithmetic and logic operations included in the CPU instruction set

Instruction set

CPU has built-in ability to execute a particular set of machine instructions, called its *instruction set*

Most CPUs have 200 or more instructions (such as add, subtract, compare, etc.) in their instruction set

CPUs made by different manufacturers have different instruction sets

Manufacturers tend to group their CPUs into “families” having similar instruction sets

New CPU whose instruction set includes instruction set of its predecessor CPU is said to be *backward compatible* with its predecessor.

Registers

- Special memory units, called registers, are used to hold information on a temporary basis as the instructions are interpreted and executed by the CPU.
- Registers are part of the CPU (not main memory) of a computer.
- The length of a register, sometimes called its *word size*, equals the number of bits it can store.
- With all other parameters being the same, a CPU with 32-bit registers can process data twice larger than one with 16-bit registers.

Types of registers

S.No	Name of the register	Function
1	Memory Address(MAR)	Holds address of the active memory location
2	Memory Buffer(MBR)	Holds contents of the accessed (R/W) memory word
3	Program Control(PC)	Holds address of the next instruction to be executed
4	Accumulator(A)	Holds data to be operated upon, intermediate results, and the result
5	Instruction(I)	Holds an instruction while it is being executed
6	Input/Output(I/O)	Used to communicate with the I/O devices

Processor speed

- Computer has a built-in *system clock* that emits millions of regularly spaced electric pulses per second (known as *clock cycles)*
- It takes one cycle to perform a basic operation, such as moving a byte of data from one memory location to another
- Normally, several clock cycles are required to fetch, decode, and execute a single program instruction
- Hence, shorter the clock cycle, faster the processor.
- Clock speed (number of clock cycles per second) is measured in Megahertz (106 cycles/sec) or Gigahertz (109 cycles/sec)

Processor and memory

- Internal structure of processor
  - Memory structure
  - Determining the speed of a processor
- Different types of processors available
  - Determining the capacity of a memory
- Different types of memory available
- Several other terms related to the processor and main memory of a computer system

Types of processor

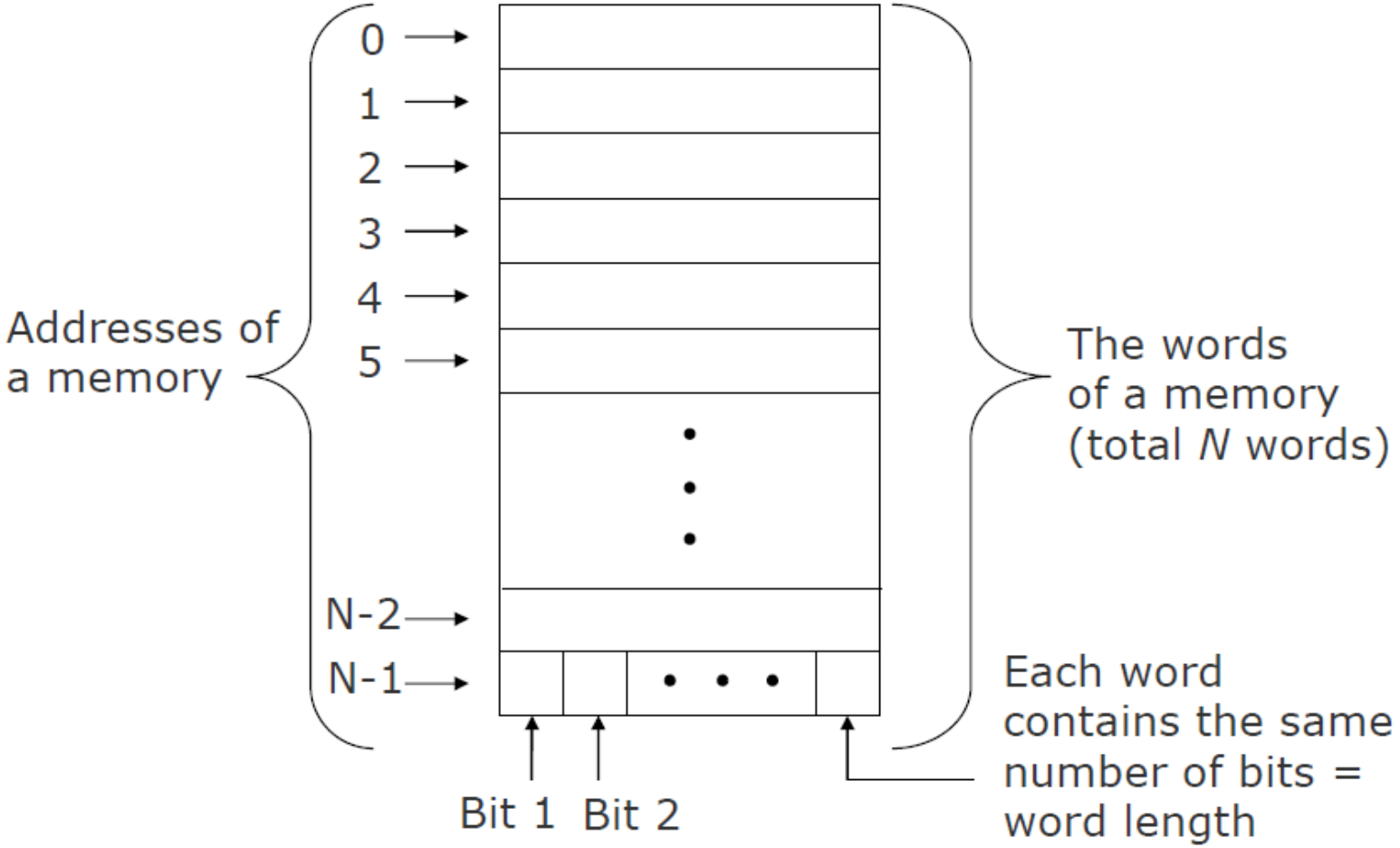
Types of architecture	Features	Usage
CISC (Complex instruction set computer)	• Large instruction set • Variable-length instructions • Variety of addressing modes • Complex & expensive to produce	personal computers
RISC (Reduced Instruction set Computer)	• Small instruction set • Fixed-length instructions • Reduced references to memory to retrieve operands	Workstations

Types of architecture	Features	Usage
EPIC (Explicitly Parallel Instruction Computing)	Allows software to communicate explicitly to the processor when operations are parallel • Uses tighter coupling between the compiler and the processor • Enables compiler to extract maximum parallelism in the original code, and explicitly describe it to the processor	Mostly used in high-end servers and workstations
Multi-Core Processor	Processor chip has multiple cooler-running, more energy efficient processing cores • Improve overall performance by handling more work in parallel • can share architectural components, such as memory elements and memory management	Mostly used in high-end servers and workstations

Memory-hierarchy(Main memory)

- Every computer has a temporary storage built into the computer hardware
  - It stores instructions and data of a program mainly when the program is being executed by the CPU.
- This temporary storage is known as main memory, primary storage, or simply *memory*.
- **Memory vs. storage :** There are some distinct and important differences
  - Memory refers to the location of short-term data
  - Storage refers to the location of data stored on a long-term basis.
- Physically, it consists of some chips either on the motherboard or on a small circuit board attached to the motherboard of a computer
  - It has random access property and is volatile.

Main memory organization



Types of memory

Primary memory(main memory)

- RAM
- ROM

Secondary Memory

- Hard Disk
- Compact Disc
- Flash disk

Cache memory

Primary memory

When a program or the data is activated for execution, the processor loads instructions from the secondary memory into the main memory and then starts execution.

It is a volatile memory due to which any unsaved data is lost when a power cut occurs

Primary memory is of two types: RAM and ROM.

RAM: Random Access Memory (RAM)

A memory scheme responsible for storing data on a temporary basis.

Can be promptly accessed by the processor as and when needed. It is volatile in nature.

RAM stores data randomly and the processor accesses these data randomly from its storage.

RAM is considered "random access" because you can access any memory cell directly if you know the row and column that intersect at that cell.

DRAM - Dynamic Random-Access Memory used for dynamic data storage. A cell(transistor and a capacitor) in DRAM consists of one-bit information. This capacitor and transistor are extremely small in size.

SRAM full form is Static Random-Access Memory. This type of RAM stores static data in memory which remains active until there is a power supply.

ROM - Read Only Memory

It is a permanent form of storage & stays active regardless of whether power supply to it is turned on or off. ROM devices do not allow data stored on them to be modified.

- **MROM:** Here data is pre-configured via integrated circuit manufacture during the time of manufacturing. Due to this, the user cannot change the instruction stored within the MROM chip.
- **PROM:** allows writing any information/program only once using a special PROM programmer or burner device.
- **Flash ROM:** This ROM is used for transferring data between computer and digital devices.
- **EPROM:** Data can be erased as well as reprogrammed only once and stores data for a minimum of 10-20 yrs. To erase and reprogram EPROM, the user needs to pass UV light for 40 mins. Post this, the data can be recreated.
- **EEPROM:** Electrically Erasable and Programmable Read Only Memory is an electrically erasable and programmable ROM. This allows data to be erased using a high-voltage electrical charge. After this, it can be reprogrammed up to thousands of times.

Secondary memory

- Stores data and programs permanently : its retained after the power is turned off
- **Hard drive (HD):** A hard disk is part of a unit, often called a "disk drive," "hard drive," or "hard disk drive," that store and provides relatively quick access to large amounts of data on an electromagnetically charged surface or set of surfaces.
- **Optical Disk:** an optical disc drive (ODD) is a disk drive that uses laser light as part of the process of reading or writing data to or from optical discs. Some drives can only read from discs, but recent drives are commonly both readers and recorders, also called burners or writers.

- Compact discs, DVDs, and Blu-ray discs are common types of optical media which can be read and recorded by such drives.

- Optical drive is the generic name; drives are usually described as "CD" "DVD", or "Bluray", followed by "drive", "writer", etc.
- There are three main types of optical media: CD, DVD, and Blu-ray disc.
- CDs can store up to 700 megabytes (MB) of data DVDs can store up to 8.4 GB of data.
- Blu-ray discs, which are the newest type of optical media, can store up to 50 GB of data.
- This storage capacity is a clear advantage over the floppy disk storage media (a magnetic media), which only has a capacity of 1.44 MB.
- A storage module made of flash memory chips. A Flash disks have no mechanical platters or access arms, but the term "disk" is used because the data are accessed as if they were on a hard drive. The disk storage structure is emulated.

Cache memory

- It is commonly used for minimizing the memory processor
- speed mismatch.
- It is an extremely fast, small memory between CPU and main memory whose
- access time is closer to the
- processing speed of the CPU.
- It is used to temporarily store very active data and instructions during processing

Software

- An organized collections of computer data and instruction.
- There are two major categories:
  - system software that provides the basic non-task-specific functions of the computer,
  - application software which is used by users to accomplish specific tasks.

Comparison between Main memory (RAM) and Secondary Memory (Hard disk)

RAM	Hard Disk (Hard Drive)
Memory	Storage
Smaller amount (typically 500 MB-6 GB)	Much larger amount (typically 80GB to 1000 GB)
Temporary storage of files and programs	Permanent storage of files and programs
A little like your real desktop - has only your current work on it (which could be ruined by a spill of Coke or coffee!)	Like a file cabinet - has long-term storage of work (it's safe from spills!)
Contents disappear when you turn off power to the computer and when the computer crashes	Contents remain when you turn off the power to the computer (they don't disappear unless you purposely delete them), and when the computer crashes
Consists of chips (microprocessors)	Consists of hard disks (platters)
When you want to use a program, a temporary copy is put into RAM and that's the copy you use	Holds the original copy of the program permanently

Systems Software

- It is responsible for controlling, integrating, and managing the individual hardware components of a computer system.
- Software and the users of the system see it as a functional unit without having to be concerned with the low-level details Ex: transferring data from memory to disk, or rendering text onto a display.
- Generally, system software consists of an operating system and some fundamental utilities .
- Ex: disk formatters, file managers, display managers, text editors, user authentication (login) and management tools, and networking and device control software.

Applications software

It is used to accomplish specific tasks other than just running the computer system.

Application software may consist of a single program, such as an image viewer;

A small collection of programs (often called a software package) that work closely together to accomplish a task, such as a spreadsheet or text processing system;

A larger collection (often called a software suite) of related but independent programs and packages that have a common user interface or shared data format.

Ex: Microsoft Office - which consists of closely integrated word processor, spreadsheet, database, etc.;

A software system, such as a database management system - is a collection of fundamental programs that may provide some service to a variety of other independent applications.

Comparison Application Software and System Software

	System Software	Application Software
	Computer software, or just software is a general term primarily used for digitally stored data such as computer programs and other kinds of information read and written by computers. App comes under computer software though it has a wide scope now.	Application software, also known as an application or an "app", is computer software designed to help the user to perform specific tasks.
Example:	1) Microsoft Windows 2) Linux 3) Unix 4) Mac OSX 5) DOS	1) Opera (Web Browser) 2) Microsoft Word (Word Processing) 3) Microsoft Excel (Spreadsheet software) 4) MySQL (Database Software) 5) Microsoft PowerPoint (Presentation Software) 6) Adobe Photoshop (Graphics Software)
Interaction:	Generally, users do not interact with system software as it works in the background.	Users always interact with application software while doing different activities.
Dependency:	System software can run independently of the application software.	Application software cannot run without the presence of the system software.